Ambarish Sarkar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.16
GAZOLE H.N.M.HIGH SCHOOL (H.S.)	Class XII (West Bengal Council of Higher Secondary Education)	2021	91.80%
GAZOLE H.N.M.HIGH SCHOOL (H.S.)	Class X (The West Bengal Board of Secondary Education)	2019	97.00%

M.TECH THESIS/RESEARCH EXPERIENCE

Occupancy Attacks on Multi-Bank and Hybrid Last-Level Caches | M.Tech Thesis

- Investigating occupancy-based cache side-channel attacks on multi-bank and hybrid LLC architectures under different cache organizations and attacker configurations.

- Extended a cache simulation framework to support configurable multi-bank LLC designs and evaluate single-bank and multi-bank attack scenarios.

- Analyzing the security implications of randomized/secure cache architectures and their resistance to occupancy-based information leakage.

Technologies: C/C++, Computer Architecture, Cache Security, Linux, Cache Simulation

PROJECTS

Detection and Localization of Network Middleboxes

- Developed a Mininet-based testbed to detect network middleboxes using differential packet-loss behavior between specially generated traffic flows.

- Emulated traffic shapers, compression middleboxes and strict-priority queues using Linux tc/HTB, NetfilterQueue, iptables and DSCP-based classification.

- Automated experiments across single and chained middlebox topologies, identifying cases where chained middleboxes mask the observable detection signal.

- Explored TTL-based localization using Scapy and ICMP Time Exceeded responses and analyzed its limitations in controlled network topologies.

Technologies: Python, Mininet, Scapy, Linux Networking, NetfilterQueue, iptables, tc/HTB

Secure Peer-to-Peer Transaction Platform

- Developed a web-based peer-to-peer transaction platform supporting user authentication, fund transfers, transaction history and profile management.

- Implemented security mechanisms including bcrypt password hashing, prepared SQL statements, CSRF protection, session management and brute-force protection.

- Used MySQL transaction handling for fund-transfer operations and containerized the application using Docker and Docker Compose.

Technologies: PHP, MySQL, JavaScript, HTML/CSS, Docker, Docker Compose, Web Security

Smart Attendance System using Face Recognition

- Developed a web-based attendance management system using facial recognition to register students and record attendance through webcam input.

- Integrated face detection and facial embeddings using face-api.js and a Python face-recognition service for similarity-based identity matching.

- Built authentication, student and subject management, and attendance persistence using React, TypeScript and Supabase.

Technologies: React, TypeScript, Python, Flask, Supabase, face-api.js, Computer Vision

SKILLS

Programming Languages: C, C++, JavaScript, SQL

Web & Databases: React, MySQL, Supabase, HTML/CSS

Systems & Networking: Linux, Mininet, Scapy, Wireshark, iptables, NetfilterQueue, tc/HTB

Tools & Technologies: Git, GitHub, Docker, Docker Compose

Technical Areas: Computer Architecture, Cache Security, Computer Networks, Data Structures and Algorithms

POSITIONS OF RESPONSIBILITY

Teaching Assistant - Advanced Data Structures and Algorithms | IIT Hyderabad

- Assisting in the delivery of the Advanced Data Structures and Algorithms course for undergraduate/postgraduate students.

- Supporting students with algorithmic concepts, problem solving and course-related doubts, along with evaluation of coursework and assignments.

RELEVANT COURSEWORK

- **Algorithms:** Advanced Data Structures and Algorithms

- **Networking:** Advanced Computer Networks, Network Engineering, Networked Wireless Systems

- **Security:** Cryptography, Network Security